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{
  "questions": [
    {
      "question": "Which of the following is the lightest and most abundant chemical element in the universe?",
      "options": ["Hydrogen", "Oxygen", "Nitrogen", "Carbon"],
      "correct_answer": 1,
      "solution": "Hydrogen is the most abundant chemical element, making up about 75% of the universe's elemental mass. Hence, Option (1) is the right answer."
    },
    {
      "question": "In the context of the kinetic theory of gases, which of the following properties is NOT assumed to be a characteristic of an ideal gas?",
      "options": ["Particles do not exert forces on each other except during collisions.", "Particles occupy negligible space.", "Particles have varying masses.", "All collisions are perfectly elastic."],
      "correct_answer": 4,
      "solution": "The kinetic theory assumes that all particles of an ideal gas have identical mass. The statement that particles have varying masses is incorrect. Hence, Option (4) is the right answer."
    },
    {
      "question": "If a data set has values 3, 7, 7, 2, 5, 7, what is the mode of this data set?",
      "options": ["7", "5", "6", "3"],
      "correct_answer": 1,
      "solution": "The mode is the number that appears most frequently in a data set. Here, the number 7 appears most frequently. Hence, Option (1) is the right answer."
    }
  ]
}
```